

Snowflake CoCo CLI Enterprise SupplyGuard Agent

Team: **EquiSaaS BD** | Problem Statement: **Intelligent Workflow Automation Agents**

Role	Member Name	Email Address
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Executive Summary:

Global supply chains suffer millions in losses due to unmonitored supplier delays, maritime congestion, and inventory stockouts. Traditional ERP/BI systems are passive: they log anomalies long after damage is done.

Cortex-SupplyGuard is an autonomous cognitive resilience system powered by **Snowflake CoCo CLI Agent Skills**. It continuously scans warehouse tables, synthesizes unstructured incident logs using Snowflake Cortex AI, and executes policy-bounded transactional mitigations in under 3 seconds.

1. Problem Brief & Business Domain

Target Persona: VP of Supply Chain, Operations Managers, Procurement Directors

- **Real Business Problem:** Unexpected lead-time spikes and port delays cause assembly line stoppage. Traditional BI tools alert only after line halt occurs.
- **Current Pain Point:** Operational teams spend 2 to 5 days manually cross-referencing SQL inventory tables with unstructured logistics email reports.
- **Our Solution (Cortex-SupplyGuard):** Autonomous multi-step workflow orchestrating 3 specialized CoCo Agent Skills to detect statistical anomalies, deduce root cause via Snowflake Cortex AI, and issue policy-bounded emergency purchase orders.

Key Business Metric	Traditional Manual Process	Autonomous CoCo Agent
Resolution Time (MTTR)	2 - 5 Days of manual triage	Instant (< 3 Seconds)
Stockout Prevention	Reactive after production halt	100% Proactive Buffer Reroute
Quantified Loss Avoided	High financial leakage	\$200,000+ per incident
Governance & Policy	Prone to manual oversight	100% Auditable Ledger

2. Modular CoCo Agent Skills Architecture

Built using standardized SKILL.md CoCo specification format

Skill Name & Directory	Role & Capabilities	Input / Output Schema
data-anomaly-detector
Skills/ai/data_anomaly_detector/	• Automates data quality checks on user data state for 1000 records in real-time delay 100ms for raw data analysis table Composite Risk Score (0-100)	
cortex-reasoner
Skills/ai/cortex_reasoner/	• Snowflake AI Cortex / ai_cortex_reasoner for text reasoning • Synthesizes unstructured data into structured incidents (errors, incidents, trends) • Root Cause Analysis	
action-executor
Skills/ai/action_executor/	• Enforces policy actions (e.g. \$50k limit approve) • Issues emergency POs 10 seconds for stock policy budgets 10 audit 10 k	

3. Production Deployment & Deliverables

Fully deployed for free on Vercel Global Serverless Cloud Infrastructure

- **Live Production Web Application:**

<https://cortex-supplyguard-agent.vercel.app>

- **Public GitHub Repository:**

<https://github.com/kholipha-ahmmad-al-amin/cortex-supplyguard-agent>

- **Automated Test Suite Status:** 100% Passing (6/6 PyTest suite verified)

EquiSaaS BD Team Members:

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